NERAM · PERIOPERATIVE THERMAL DIGITAL TWIN

NERAM Viewer User Guide

What every screen, number and graph means, and how to drive a replay from start to finish.



Proof-of-concept. Everything shown is a **replay** of pre-computed data. Values marked SIMULATED or ASSUMED are not measurements. The temperature is a *patient temperature signal* whose body site is not established. It is not core temperature. Not for treatment decisions.

For: Sai (NERAM team) · Viewer: NERAM Thermal Twin Viewer (Unity 6000.3.26f1, build of 18 Oct 2026) · Screenshots from the Unity Editor at 1366 × 768.

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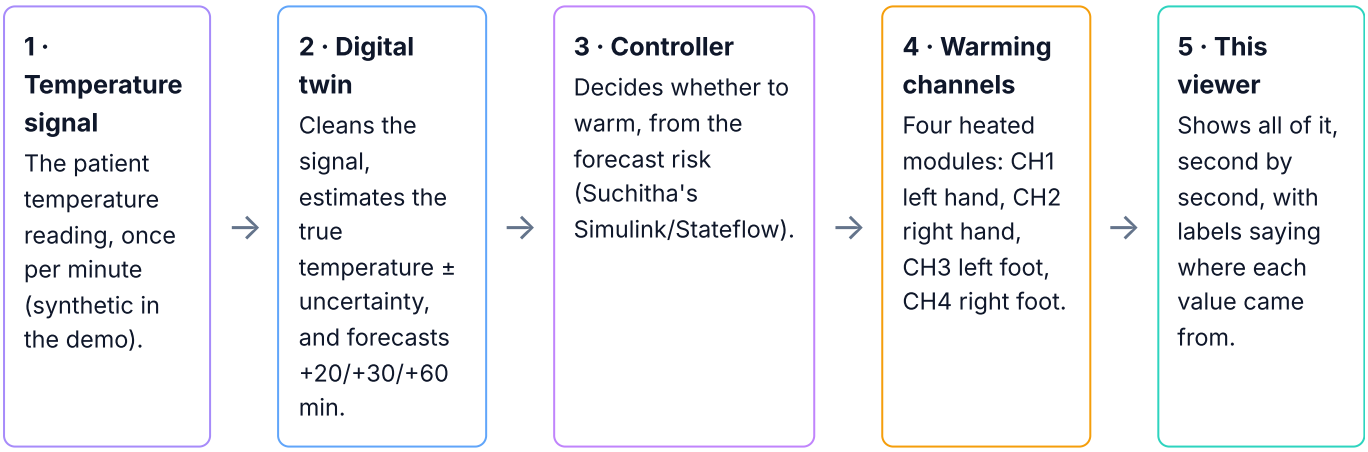
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1 The big picture

The app is a **replay viewer**. A replay file holds one value per second for a whole operation. The viewer plays it back like a video and draws it in 3D and on charts. It computes no temperatures and makes no decisions itself. Everything you see was produced earlier by the NERAM pipeline:



The idea in one sentence: instead of waiting for the patient to become cold, the twin predicts that they will be cold in 20–30 minutes, and the controller starts gentle warming early ("predictive warming").

> **Words you will see everywhere**
> **T_obs** = the observed temperature reading · **T_est** = the twin's estimate of the true temperature · **SD** = the uncertainty of that estimate · **t_hypo** = the hypothermia line · **t_target** = the warming target · **Heat %** = how hard a channel is commanded to heat (0–100 %) · **T+hh:mm:ss** = time since the replay started.

2 Quick start: play a replay in 60 seconds

1. **Open the app** from the taskbar icon (NERAM.exe). It opens as a normal window. Press `F11` or **FULLSCREEN** for fullscreen.
2. **Pick a scenario:** press ☰ **MENU** → **Scenarios**, scroll to a card and press **Load**. The app starts with *Predictive warming (SYN_SCN_B)*, which is the best one to show.
3. **Press PLAY** (bottom left) or `Space`. At 1× one second of replay takes one real second, and a whole operation is about 2 hours.
4. **Speed it up:** press **60×** in the SPEED group (or key `5`). Two hours then play in about 2 minutes.
5. **Jump to the interesting moments** with the JUMP buttons: **Transition** (next controller state change), **Fault**, **Safety**, **Clinician**, **Flag**. Or click anywhere on the coloured **timeline** strip.
6. **Switch screens** at the top: **THEATRE** (3D room), **MONITOR** (big numbers and graphs), **FULL ANALYSIS** (everything together).

Keyboard shortcuts

Key	Does	Key	Does
`Space`	Play / pause	`M`	Open the MENU
`1` … `5`	Speed 1× · 2× · 4× · 10× · 60×	`N`	MONITOR screen
`←` `→`	Step one frame back / forward	`D`	Diagnostics
`T` `F` `S`	Jump to next Transition / Fault / Safety event	`R`	Free Roam camera (walk around)
`C`	Next camera preset	`P`	Presentation mode (hides chrome)
`F11`	Fullscreen on / off	`Esc`	Close the open drawer / deselect

3 Top bar, playback bar and footer

Where	What it tells you
Scenario name	ID (e.g. **SYN_SCN_B**; "SYN_" = synthetic) and title of the loaded replay.
Chips under it	SYNTHETIC = not patient data. PLACEHOLDER = a scripted stand-in replay. FAULT TEXT TBD = fault explanations not delivered yet. Hover a chip for the full meaning.
PAUSED / PLAYING ×60	Playback state and speed.
Ctrl … · Twin …	Controller state and twin validity right now.
T+00:45:54	Replay clock: time since the start of the case. Below it: frame number and elapsed time.
Timeline strip	The whole case as colours: grey normal · purple predictive warming · orange active warming · green target maintenance · blue clinician warming · yellow twin degraded · red safety shutdown. Ticks mark events. Click or drag to seek.
Footer	The disclaimer, always visible, plus cohort, schema and parameter versions on the right.

4 THEATRE: the 3D operating room



SYN_SCN_B at T+00:45:54: predictive warming, all four channels at 33 %.

The patient and the four modules. Mitts on both hands (CH1 left, CH2 right) and boots on both feet (CH3 left, CH4 right). Their **orange glow follows the replayed heat command**: no glow at 0 %, brighter at higher %. It shows the command, not a measured skin temperature.

Floating labels. "CH1 33 %" is the channel and its heat command; it reads "off" at 0 %, and "SHUTDOWN", "FAULT" or "DISCONNECTED" when relevant. "CTRL PREDICTIVE_WARMING" over the bedside controller is the controller state.

VIEW buttons. External = normal look. **Internal** = see through the shell to the heater layers. **Thermal** = colour the modules by heat. **Exploded** = pull a module apart into its layers (liner, sensors, heater, insulation, shell).

CAMERA. Inspect = orbit with the mouse (drag to rotate, right-drag to pan, wheel to zoom). **Free Roam (R)** = walk around: WASD to move, mouse to look, Space/Ctrl up/down, Shift faster, Esc to leave. **Preset** = jump to saved views. **Reset View** returns to the start.

SELECT A REGION (right). Click a hand, foot or the bedside controller to see its details: command, status, the two module temperature sensors (A and B), connection.

Clickable screens. The labels "MONITOR · VALUES", "TEMPERATURE GRAPHS" and "PIPELINE · FORECAST" on the theatre monitors open those MONITOR pages.

Amber badge "SYNTHETIC STAND-IN DATA · NOT NERAM TWIN / CONTROLLER OUTPUT". It appears on the demo replays built by the scripted generator, so they cannot be mistaken for real NERAM output.

5 MONITOR: the four pages

MONITOR shows the same moment in large type, like a bedside monitor. Use the four buttons at the top (VALUES, TEMPERATURE GRAPHS, CHANNEL GRAPH, PIPELINE · FORECAST). ◂ **THEATRE** and **FULL ANALYSIS** switch screens; **Close** goes back.

5.1 VALUES: the current numbers



Panel	How to read it
CONTROLLER	Big coloured state (here **PREDICTIVE WARMING**), when it started ("since T+00:21:00"), **why** ("reason: Forecast risk (+30 min)") and any alert chip ("ALERT: FORECAST RISK AT +20/+30 MIN").
TEMP. SIGNAL	The latest reading **T_obs** (36.12 °C). Purple tag OBS SIMULATED = synthetic.
TWIN ESTIMATE	The twin's best estimate **T_est** (36.13 °C) with its uncertainty (± 0.050 °C = 1 SD), and whether the twin is **Valid**. It can differ from the reading, because the twin ignores glitches.
WARMING CHANNELS	One row per limb: **Status** (HEATING / off / SHUTDOWN / FAULT), **Cmd %** = heat command, **T_A** and **T_B** = the module's two temperature sensors (simulated hardware).
CONTROLLER · ALERTS · FAULTS	State, alert, fault code (none) and whether a safety shutdown is active.
SIGNAL QUALITY	Is there a reading now? How much was missing in the last 10 min? **P(outlier)** = the twin's probability that the reading is an artefact. **P(level shift)** = probability of a sudden real jump. **Twin data stale** = whether the twin output is out of date. These are model probabilities, not confirmed facts.

5.2 TEMPERATURE GRAPHS: the two most important charts



Chart A · Patient temperature signal (whole case). The raw reading over the entire operation (here 2 h 10 min). Time runs left to right (0:00 → 2:00). Grey gaps would mean "no reading" and are never filled in. The **spike near 1:10** is a deliberate artefact in the synthetic data, and the **drop near 1:45** a level shift; the twin is built to handle both.

Chart B · Measured vs Twin (±60 min around now). The white vertical line is **now**. Left of it is the last hour; right of it (purple area) is the **forecast**.

In chart B	Means
White dots	Readings T_obs, one per minute
Blue line + band	Twin estimate T_est ± 1 SD
Green dashed	t_target (warming target)
Red dashed	t_hypo (hypothermia line)
Purple bars	Forecast at +20, +30, +60 min: thick = 80 % range, thin = 95 % range
Orange / pink marks	"artefact?" / "level shift?" flags

> **How to read the forecast:** if the purple bars sit on or below the red t_hypo line, the patient is predicted to become hypothermic. That is what triggers **predictive warming** before it happens. The label "No-intervention forecast" means the prediction assumes no warming.

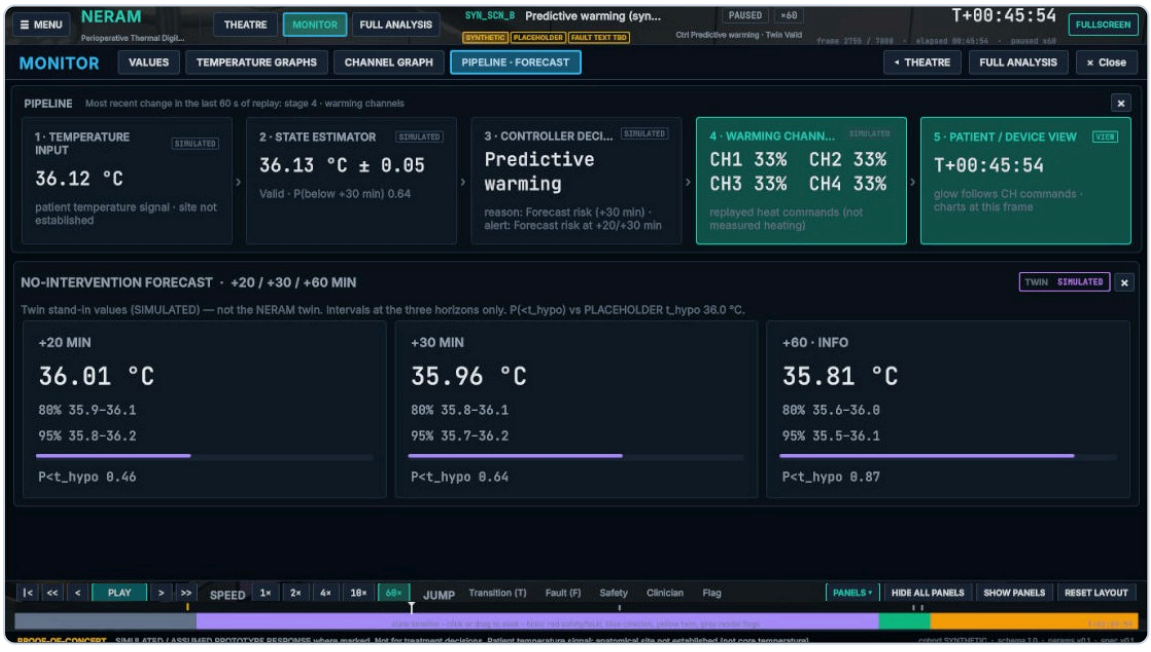
Toggles under chart B: **±SD** shows or hides the band, **Flags** the artefact marks, **ASSUMED compare** overlays the assumed warming response, **Auto** rescales the y-axis. The labels "PLACEHOLDER (TBD)" next to t_target and t_hypo mean those lines use stand-in values until the real ones are set (see section 9).

5.3 CHANNEL GRAPH: what the heaters were told to do



One lane per channel (teal = hands, amber = feet) showing the **heat command 0–100 %** over the whole case. Here warming starts at about 0:21, holds around 30–35 %, pauses briefly near 1:42 and resumes. The numbers bottom right (CH1 33 % …) are the values at the current time. These are **commands from the controller log, not measured heating**.

5.4 PIPELINE · FORECAST: the whole chain in one row



Pipeline (top): the five steps from section 1, left to right: 1 temperature input → 2 state estimator (twin estimate ± SD and P(below +30 min)) → 3 controller decision with its reason → 4 warming channel commands → 5 what the 3D view shows. The **highlighted card** is the step that changed most recently.

Forecast cards (bottom): for +20, +30 and +60 minutes: the predicted mean temperature, the 80 % and 95 % ranges, and **P<t_hypo** = the probability of being below the hypothermia line then (the bar shows it). The +60 card is marked **INFO** because the controller only acts on +20/+30. The forecast is **hidden whenever the twin is not VALID**.

6 FULL ANALYSIS: everything at once



Left column: the controller banner, temperature signal and twin estimate, the channel table and signal quality. It is the VALUES page in compact form; scroll it for more.

Top strip: the five-step pipeline (section 5.4).

Centre: the live 3D theatre with the VIEW and CAMERA buttons.

Right column: the selected region (click a limb) and the **forecast** cards with P<t_hypo bars.

Bottom row: chart A (whole-case signal), chart B (measured vs twin ±60 min with forecast), chart C (channel commands).

Every panel has a source tag top right, and an **×** to hide it. **PANELS ▾**, **HIDE ALL PANELS**, **SHOW PANELS** and **RESET LAYOUT** (bottom right) bring them back.

> Use FULL ANALYSIS for explaining and judging; use THEATRE and MONITOR for a clean demo video.

7 Controller states, twin validity and safety shutdown

Controller state	What it means
NORMAL	Monitoring only; no warming.
PREDICTIVE_WARMING	The forecast says the patient will probably become hypothermic within 20–30 min, so gentle warming starts early (capped at a lower heat).
ACTIVE_WARMING	The estimate is already below the hypothermia line; stronger warming.
TARGET_MAINTENANCE	Target reached; heat is trimmed to hold it.
CLINICIAN_WARMING	A clinician pressed START; warming is under manual supervision.
TWIN_DEGRADED	The twin is unreliable (too many missing readings or too much uncertainty). Amber "Twin unreliable — monitoring only"; the forecast is hidden; no automatic warming.
SAFETY_SHUTDOWN	A safety fault (e.g. a module too hot). All heating forced off and **latched** until a clinician RESET.

Twin validity: No data → **Initialising** (first 20 minutes, so no forecast yet) → **Valid** (at least 10 of the last 15 minutes observed and uncertainty small enough) → or **Degraded**.

Why did it change? The "reason" after each state change: Forecast risk (+20/+30 min), T_est below t_hypo, Clinician START/STOP/RESET, Target reached/lost, Safety fault, Twin degraded/recovered.

Example: safety shutdown (scenario SYN_TC07)

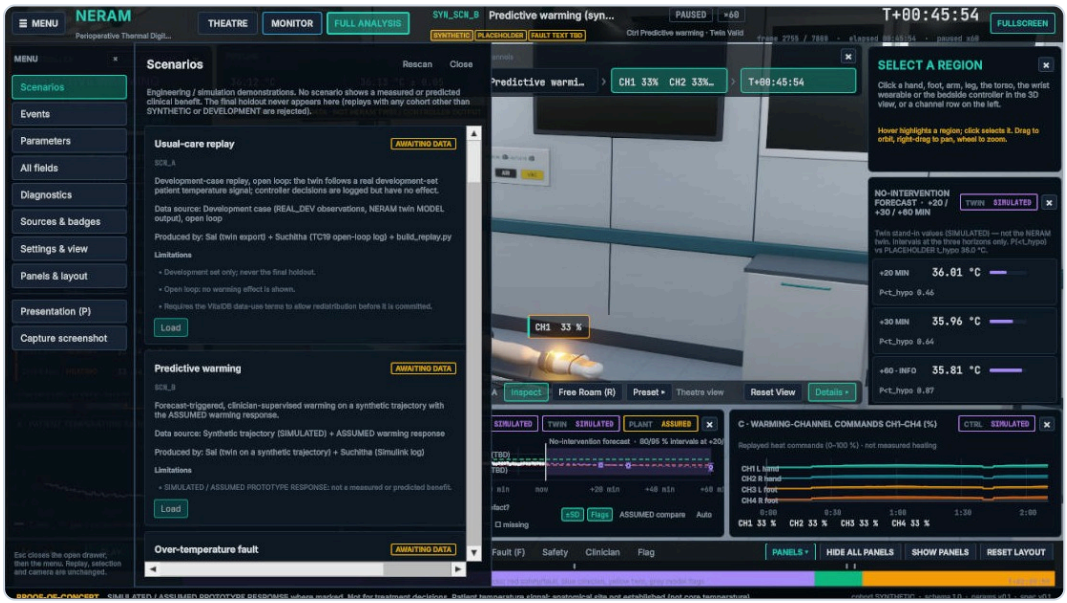


Suchitha's TC07 controller test: channel 2's module temperature reaches 41.0 °C at T+00:03:20. Red banner "SAFETY SHUTDOWN — LATCHED", fault code 206 (CH2 software over-temperature), all four channels SHUTDOWN, "Latched — awaiting clinician RESET". The log ends before any reset.

8 The MENU

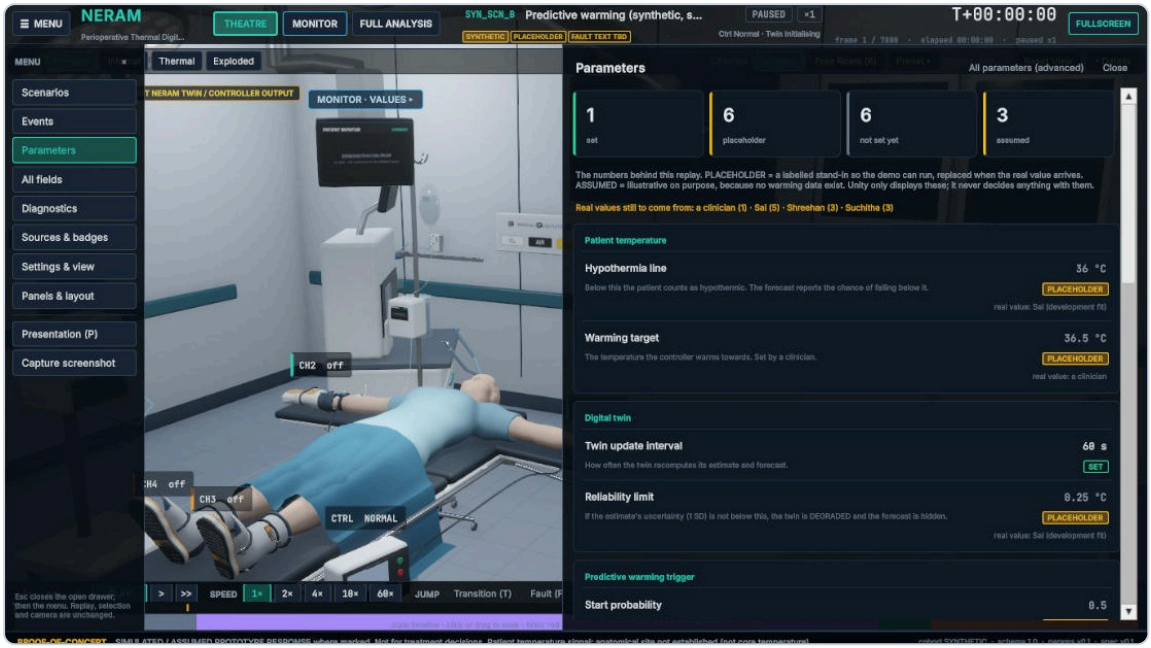
Press ☰ **MENU** (top left) or `M`. Each item opens a drawer on the right; `Esc` closes it. Opening a drawer never changes the replay time or camera.

Item	Use it to
Scenarios	Load a replay. Each card says what it shows, where the data come from, and its limitations. AWAITING DATA = a planned scenario whose real data do not exist yet.
Events	A list of every event (state changes, alerts, flags). Click one to jump there; the inspector shows the exact values at that second.
Parameters	The key numbers behind the replay (below).
All fields	Every raw value of the current second (for checking).
Diagnostics	Is the file OK, where the data come from, what is still a placeholder (below).
Sources & badges	Explains the colour tags.
Settings & view	Taller charts, UI options.
Panels & layout	Show or hide individual panels.
Presentation (P)	Clean view for a video or projector.
Capture screenshot	Saves a PNG of the screen.



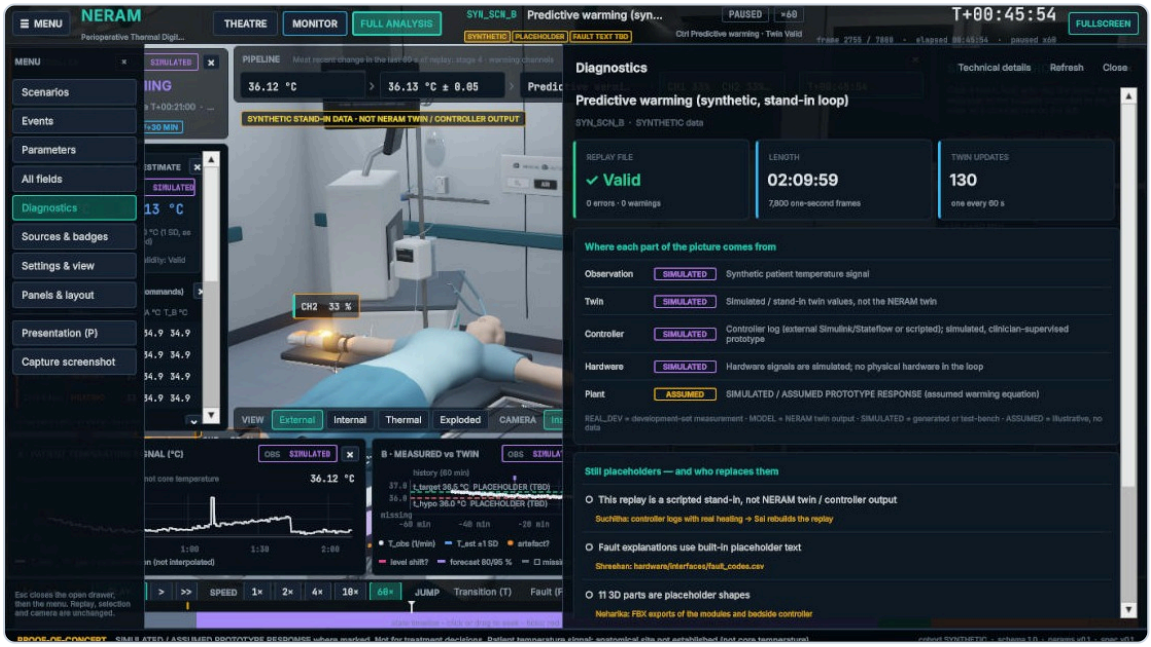
The Scenarios drawer: the planned cases SCN_A–D at the top (awaiting real data), and the playable SYN_ replays further down.

Parameters: the key numbers, in plain words



Four tiles count the key parameters: **set** (a real agreed value, e.g. the twin updates every 60 s), **placeholder** (a labelled stand-in so the demo can run), **not set yet** (no value at all yet), and **assumed** (illustrative on purpose, because no warming data exist). Each row says what the number does, its current value, and **who supplies the real value**: you (development fit), a clinician (warming target), Suchitha (controller limits) or Shreehan (hardware safety limits). **All parameters (advanced)** opens the complete audit table.

Diagnostics: is everything OK?



Three tiles: **Replay file ✓ Valid** (0 errors, 0 warnings, using the same rules as the Python validator), **Length** (2 h 10 min = 7,800 one-second frames), **Twin updates** (130, one per minute). Then **where each part of the picture comes from**, and a checklist of **what is still a placeholder and who replaces it**. **Technical details** shows file paths, hashes, schema and performance.

9 Colours, tags and "placeholder" labels

Tag	Means
REAL_DEV	A real measurement from the VitalDB development set (none in the demo yet: that needs the development fit).
MODEL	Output of the real NERAM digital twin (e.g. SYN_TC07's twin).
SIMULATED	Generated data or a test bench: synthetic temperature, stand-in twin, controller log, simulated hardware.
ASSUMED	An illustrative assumption with no data behind it: the warming effect on the patient.

Why do I still see "placeholder"?

Because some inputs really are stand-ins, and the app says so honestly. Each label goes away when its real input arrives:

Label	Goes away when
Amber badge + PLACEHOLDER chip	Suchitha sends controller logs with real heating; the replays are then rebuilt from the real NERAM twin and her controller (SYN_TC07 already is).
FAULT TEXT TBD	Shreehan delivers *fault_codes.csv* (the fault explanations).
PLACEHOLDER next to t_hypo / t_target	The development fit gives real values, and a clinician sets the warming target.
Placeholder 3D parts	Neharika delivers the CAD/FBX models of the modules and the controller.

Colours of states and charts

States: grey normal · purple predictive warming · orange active warming · green target maintenance · blue clinician warming · yellow twin degraded · red safety shutdown. Charts: red dashed = hypothermia line, green dashed = target line. Hands are teal, feet amber, in every chart.

10 Which scenario, a demo script, FAQ

Scenario	Shows	Use for
SYN_SCN_B	Predictive warming: the forecast predicts hypothermia, warming starts early, target is held	**Main demo**
SYN_SCN_D	Twin degraded: readings drop out, the forecast hides, monitoring only, then recovery	Safety of the twin
SYN_SCN_C / SYN_TC07	Over-temperature: safety shutdown and latch (TC07 = Suchitha's real controller log)	Safety of the device
SYN_SCN_A0	Layout stand-in for the usual-care case	Background only
SYN_ENG_01–06	Engineering cases (faults, disconnects, stale data)	Testing

2-minute demo script

1. THEATRE, SYN_SCN_B loaded: "Patient on the table, four warming modules, controller at the bedside. Everything is a replay; tags show what is simulated."
2. Press 60× and PLAY. At about 0:21 the gloves start to glow: "the controller started **predictive warming**."
3. Pause and press MONITOR → TEMPERATURE GRAPHS: "the purple forecast bars approach the red hypothermia line, so the twin predicted the drop."
4. PIPELINE · FORECAST: walk the five steps left to right, then the P<t_hypo bars.
5. Load SYN_TC07 → JUMP Safety: "a module over-heats, everything shuts off and stays latched."
6. Finish on the footer: proof-of-concept, simulated and assumed values marked.

FAQ

Why is there no forecast at the start? The twin needs 20 minutes of readings first (Initialising).

Is the temperature core temperature? No. It is a patient temperature signal whose body site is not established.

Does Unity decide when to warm? No. Decisions come from the controller log; Unity only replays them.

Do the gloves show real heating? They show the heat *command*. The patient's response is ASSUMED.

Something looks wrong? MENU → Diagnostics shows whether the file is valid and what is a placeholder.